

Candidate Name

Centre Number

Candidate Number

**ZIMBABWE SCHOOL EXAMINATIONS COUNCIL**

General Certificate of Education Advanced Level

**CHEMISTRY****6031/2**

PAPER 2

**NOVEMBER 2021 SESSION****1 hour 30 minutes**

Candidates answer on the question paper.

Additional materials:

Data Booklet

Electronic Calculator

**TIME** 1 hour 30 minutes**INSTRUCTIONS TO CANDIDATES**

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets [ ] at the end of each question or part question.

**FOR EXAMINER'S USE**

|              |  |
|--------------|--|
| 1            |  |
| 2            |  |
| 3            |  |
| 4            |  |
| 5            |  |
| 6            |  |
| <b>TOTAL</b> |  |

**This question paper consists of 10 printed pages and 2 blank pages.**

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Answer all questions

For  
Examiner's  
Use

- 1 (a) Outline the use of ion exchange in treatment of industrial waste.

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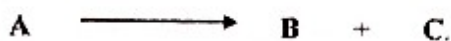
[3]

- (b) (i) Define the term *half-life*.

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- (ii) The reaction shown is a first order reaction and has a half-life of 30 seconds.



Calculate the initial rate, given that the initial concentration of A was  $1.0 \times 10^{-3} \text{ mol dm}^{-3}$ .

Initial rate = \_\_\_\_\_

[4]

- (c) Bromine exist as a mixture of two naturally occurring isotopes  $^{79}\text{Br}$  and  $^{81}\text{Br}$ . The relative atomic mass of bromine is 79.9.

Calculate the percentage abundance of each isotope.

% abundance of  $^{79}\text{Br}$  \_\_\_\_\_

% abundance of  $^{81}\text{Br}$  \_\_\_\_\_

[3]  
[Total: 10]

- 2 (a) (i) Place the organic acids  $\text{CH}_3\text{CO}_2\text{H}$ ,  $\text{C}/\text{CH}_2\text{CO}_2\text{H}$  and  $\text{Cl}_2\text{CHCO}_2\text{H}$  in order of their decreasing  $K_a$  values.

\_\_\_\_\_

- (ii) Explain the trend in (a)(i).

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

[3]

- (b) Sketch a titration curve for the titration of  $25.00\text{ cm}^3$  of  $0.1\text{ mol dm}^{-3}$  sodium hydroxide solution with  $0.05\text{ mol dm}^{-3}$  sulphuric acid to show changes in pH.

[3]

(c) When concentrated  $\text{HCl}$  is added to a pink solution of aqueous cobalt (II) ions the colour changes to blue. The original colour is generated on addition of water.

(i) Draw the 3 dimensional shape of the ion responsible for the blue colour.

(ii) Write a chemical equation that accounts for the observations.

\_\_\_\_\_

(iii) If the reaction mixture, is heated the blue colour reappears.

State with a reason, whether the reaction is exothermic or endothermic.

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

[4]

[Total: 10]

3 The chlorides of group (IV) elements exist as  $\text{XCl}_4$  or  $\text{XCl}_2$ .

(a) Describe the bonding, structure and shape present in the  $\text{XCl}_4$  molecules.

*bonding* \_\_\_\_\_

*structure* \_\_\_\_\_

*shape* \_\_\_\_\_

[3]



(b) All the  $XC\ell_4$  molecules are hydrolysed by water except one.

(i) Identify the chloride which is **not** hydrolysed by water.

\_\_\_\_\_

(ii) Explain why it is not hydrolysed.

\_\_\_\_\_

\_\_\_\_\_

(iii) Describe and explain the thermal stability of the Group (IV) tetrachlorides.

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

(iv) The +2 oxidation state becomes more stable relative to the +4 oxidation state down Group (IV) elements.

Write equations to illustrate this trend.

\_\_\_\_\_

\_\_\_\_\_

[7]  
[Total: 10]

4 (a) Define the terms

(i) *diazotisation*,

\_\_\_\_\_

\_\_\_\_\_

(ii) *coupling reaction*.

\_\_\_\_\_

\_\_\_\_\_

[2]

- (b) Fig. 4.1 shows how an organic compound, D is formed.

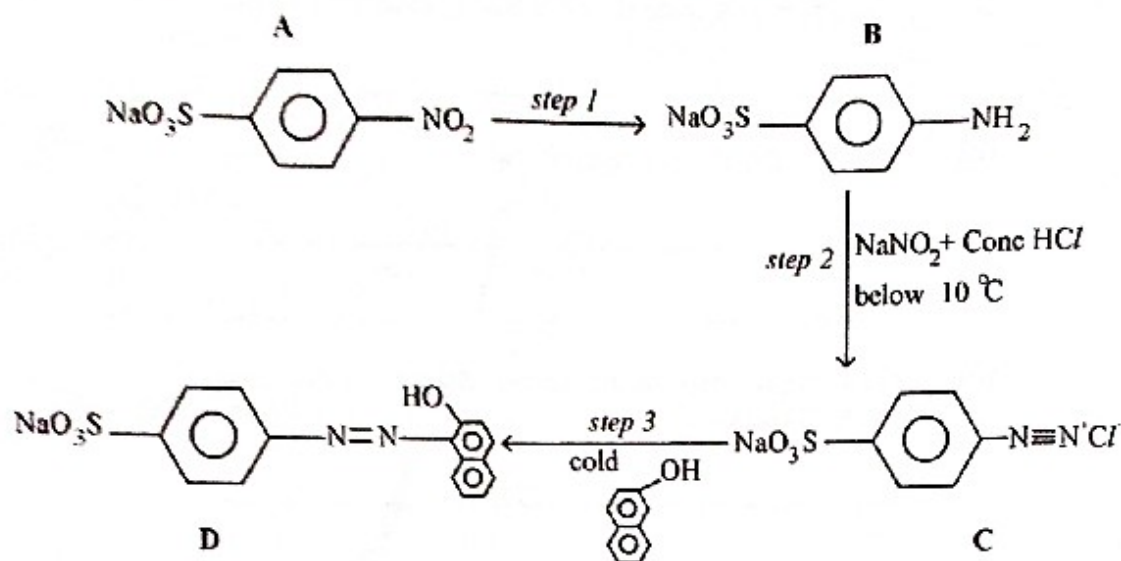


Fig. 4.1

- (i) Name the type of reaction in *step 1*.

\_\_\_\_\_

- (ii) Give the reagents and conditions for *step 1*.

reagents \_\_\_\_\_

conditions \_\_\_\_\_

- (iii) State and explain the observation made when the temperature in *Step 2* is raised above  $10^\circ\text{C}$ .

observation \_\_\_\_\_

explanation \_\_\_\_\_

- (iv) State any **one** property that makes the compound suitable for this use.

\_\_\_\_\_

[6]

- (c) Fig. 4.2 shows a section of a polypeptide.

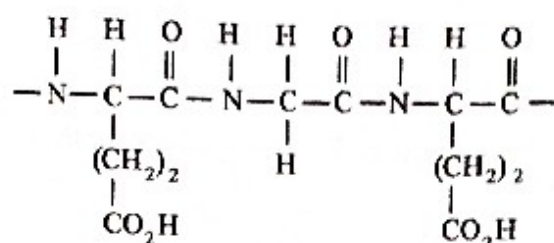
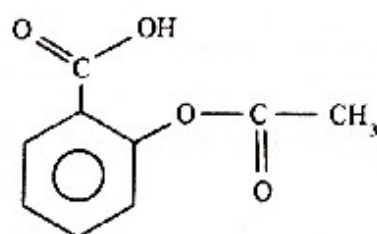


Fig. 4.2

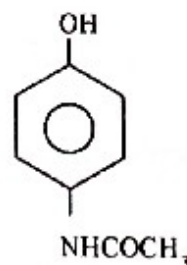
Draw structural formulae of the organic products of acid an hydrolysis of the polypeptide.

[2]  
[Total: 10]

- 5 Fig. 5.1 shows the structure of two organic compounds, M and N.



M



N

Fig. 5.1

- (a) Identify any **two** functional groups found in

(i) M, 1. \_\_\_\_\_  
2. \_\_\_\_\_

(ii) N, 1. \_\_\_\_\_  
2. \_\_\_\_\_

[4]

- (b) Suggest a reagent that can react with 2-hydroxybenzoic acid in a one step reaction to produce M.

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[1]

- (c) (i) Draw the structure of the organic product of the reaction of N and  $\text{Br}_{2(\text{aq})}$ .

- (ii) State the observations made during the reaction in (i).

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[2]

- (d) Describe and explain a reaction that can be used to distinguish M from N.

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[3]

[Total: 10]



- 6 (a) Table 6.1 shows complexes of chromium and some of their properties. When aqueous silver nitrate was added to each complex, different moles of chloride ions were precipitated per mole of complex.

Table 6.1

| molecular formula                     | colour of isomer | moles of $\text{Cl}^-$ precipitated | dipole moment |
|---------------------------------------|------------------|-------------------------------------|---------------|
| $\text{CrCl}_3(\text{H}_2\text{O})_6$ | violet           | 3                                   | zero          |
| $\text{CrCl}_3(\text{H}_2\text{O})_6$ | light green      | 2                                   | present       |
| $\text{CrCl}_3(\text{H}_2\text{O})_6$ | dark green       | 1                                   | zero          |

- (i) Explain the formation of different amounts of precipitate.
- \_\_\_\_\_
- (ii) State the type of isomerism exhibited by the complexes.
- \_\_\_\_\_
- (iii) Suggest the structure of the complex that gives two moles of chloride ions in the presence of silver nitrate.
- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_
- (iv) Explain the absence of dipole moment in the dark green complex.
- \_\_\_\_\_

[4]

(b) Iodine dissolves in both water and ether.

(i) Suggest how iodine dissolved in water could be extracted in the laboratory.

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(ii) Describe and explain how the amount of iodine extracted could be increased.

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[4]

(c) State any **two** advantages of thin layer chromatography compared to paper chromatography.

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2. 

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[2]

[Total: 10]

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